

Coding & Solution

Date	29 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the OptiCrop application's implementation. The project is developed using Python, Flask, and Machine Learning techniques to provide intelligent crop recommendations based on soil nutrients and environmental conditions. The application demonstrates clean coding practices, modular design, and reliable prediction functionality while meeting the defined project requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link/ URL	https://github.com/SaiDevi-kondari981/OptiCrop
Programming Language(s)	Python
Framework(s)Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Key Features Implemented	Crop Recommendation using Machine Learning, Soil & Climate Analysis, Prediction based on N, P, K, Temperature, Humidity, pH and Rainfall, User-friendly Flask Web Interface
Pending/Incomplete Features	No major pending features
Setup/Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , then run <code>python app.py</code> and open the Flask application in a web browser.

Code Quality Checklist

S.No	Criteria	Status(Yes/No)
1	Code is modular and organized into functions /classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes /Comments:

The OptiCrop project follows clean coding standards with a modular architecture, making the application easy to maintain and extend. Machine Learning components, data preprocessing, and Flask routes are organized into reusable modules. Proper validation and error handling improve reliability, while the project structure supports future enhancements such as fertilizer recommendation, yield prediction, and weather-based crop advisory.